

GLA UNIVERSITY, MATHURA
End Term Examination, Even Semester 2025–26
B.Tech. CSE(DA), III Year (VI Semester)
BCSE0556 – Hadoop and Big Data Analytics

Course Outcome

- CO1: Understand the concepts and challenges of big data.
CO2: Apply existing technology to collect, manage, store, query, and analyze the big data.
CO3: Apply job scheduling and resource management using Hadoop and Yarn.
CO4: Apply data summarization, query, and analysis of big data using Pig and Hive.
CO5: Design the regression model, cluster and decision tree of big data.
CO6: Experiment with hands-on experience in large-scale analytics tools to solve big data problems.

Printed Pages: 04

SET A

University Roll No.

Time: 3 Hours

Maximum Marks: 50

Instruction for students:

1. All questions should be answered into given sequenced order for all sections.
2. Answer should be brief and to-the-point with neat sketch/diagram.
3. Any missing or wrong data may be assumed to be suitable with giving proper justification.
4. Mentioned on the right-hand side margin indicates the full marks for respective questions.

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Identify the type of digital data: a. Email with subject b. Facebook post with hashtag c. CCTV Footage d. XML e. Zoom meeting recordings f. Student details table in RDBMS g. Audio books mp3s h. Big Data notes images taken from your mobile camera	4	1	An	P
2	What is Apache Spark, and how is it different from Hadoop MapReduce? Also list the key features of Apache Spark and explain why it is faster for iterative workloads.	4	2	U	F
3	Identify the type of digital data in following scenario: a. Ramesh Sundram runs a drone company and records videos using a drone camera. What will be the type of recorded footage and why? b. Vijay Deenanath Chauhan scraps data from the internet containing email exchanges between students and their supervisor at GLA University. An email contains messages with sender information, recipient information, subject lines, body text, attachments, and timestamps. What will be the type of this data and why? c. Balakrishna has collected data from Facebook page of Maha Kumbh 2025. This page has user posts containing text, a photo, a video, hashtags, mentions of other users, and a timestamp. What will be the type of this data and why? d. Rohit works in an e-commerce company. He had data of customers who placed orders. The order includes fields like customer name, address, email, order ID, product names, quantities, prices, and payment method. What will be the type of this data and why?	4	1	A	P

No.	Detail of Question	Marks	CO	BL	KL
4	Discuss the following: a. Hadoop Ecosystem b. Sqoop c. Zookeeper d. Kafka	4	1	An	P
5	Imagine you are the lead architect for a rapidly growing e-commerce platform. You are experiencing a significant increase in fraudulent transactions originating from stolen credit cards during the final checkout stage. Your goal is to design a real-time fraud detection system, using primarily Apache streaming tools (like Kafka, Flink/Spark Streaming, etc.), that can analyze transaction data as it happens (within ~500 milliseconds) to identify and block potentially fraudulent orders before payment is authorized. Outline the architecture, key components, data flow, and how stateful information (like user purchase history or velocity checks) would be managed.	4	3	U	C

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain the role of Data Scientist with one example?	3	6	C	M
7	Suppose a Node Manager crashes while running a container. What will happen to the container and how does YARN handle this failure?	3	6	A	M
8	Discuss the following: a. Hinted Hand-off b. Peer to peer networks c. NoSQL	3	5	U	C
9	Discuss the following: a. Running modes and Execution modes in Apache Pig b. Philosophy of Apache Pig c. How Pig is different from Hive?	3	4	U	C
10	Consider keyspace as 'GLA_University' with 05 Departments and running 10 Bachelor & Master Courses for their students. Write a Cassandra Query Language (CQL) for below mentioned: a) Create keyspace 'GLA_University' with basic properties. b) Create table for CSE_Dept with Primary Key having student data such as Rollno, name, course, branch, cpi. This table will contain data of CSE students. c) Display list of all tables in keyspace 'GLA_University'.	3	3	An	F

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	What is HBase? Explain HBase Architecture?	5	2	A	P

No.	Detail of Question	Marks	CO	BL	KL
12	Answer the following: a. Discuss Apache Cassandra Collection Data Types. b. Writes in Cassandra with diagram.	5	6	R	F
13	Write the HiveQL for the following: a. Create a database "bigdata". b. List down all databases. c. Show details about database "bigdata". d. Mount this newly created database "bigdata". e. Drop this database "bigdata". f. Create a managed table btech_bigdata with three columns rollno, name and gpa. g. Load data to this table "btech_bigdata" from file at location '/root/hivedemo/students.tsv'. h. Print all rows of table "btech_bigdata". i. Print only name of student from table "btech_bigdata". j. Print name and gpa of students having gpa greater than 4.8.	5	6	A	M